

PRS821S PROGRAMMING FOR SECURITY PROFESSIONAL, SEM 1, 2026
WEEK 5B LECTURE NOTES

DATABASE SECURITY

Common Database Security Issues

Databases are very attractive targets for hackers because they contain valuable and sensitive information. This can range from financial or intellectual property to corporate data and personal user data. Cybercriminals can profit by breaching companies' servers and damaging their databases in the process. Thus, database security testing is a must.

There are numerous incidents in which hackers have targeted companies handling personal customer data. Equifax, Facebook, Yahoo, Apple, Gmail, Slack, and eBay data breaches were in the news in the past few years, just to name a few. Such rampant activities underscored the need for cybersecurity software and web app testing, which aim to protect the data people share with online businesses. If these measures are implemented, hackers will be denied access to all records and documents available in online databases.

Here's a list of the top vulnerabilities commonly found in database-driven systems, along with suggestions on how to eliminate them.

No Security Testing Before Deployment

One of the most common causes of database weaknesses is negligence in the deployment stage of the development process. Although functional testing is conducted to ensure supreme performance, this type of test can't show you if the database is doing something that it is not supposed to. Thus, it is important that you test website security with different types of tests before complete deployment.

Poor Encryption and Data Breaches Come Together

You might consider the database a backend component of your setup and focus more on eliminating Internet-borne threats. It does not really work that way. There are network interfaces within the databases that hackers can easily exploit if your software security is poor. To avoid such situations, it is important to use TLS- or SSL-encrypted communication platforms.

Feeble Cybersecurity Software = Broken Database

Case in point: the Equifax data breach. Company representatives admitted that data from 147 million consumers was compromised, so the consequences are significant.

This case has proven how important cybersecurity software is to defend one's database. Unfortunately, due to a lack of resources or time, most businesses don't conduct user data security testing or provide regular patches for their systems, leaving them susceptible to data leaks.

Stolen Database Backups

There are two kinds of threats to your databases: external and internal. There are cases when companies struggle with internal threats even more than with external. Business owners can never be 100% sure of their employees' loyalty, no matter what computer security software they use or how responsible their employees seem. Anyone with access to sensitive data can steal it and sell it to third-party organisations for profit. However, there is a way to eliminate the risk: encrypt database archives, implement strict security standards, impose fines for violations, use cybersecurity software, and continuously raise your teams' awareness through corporate meetings and personal consultations.

Flaws in Features as a Database Security Issue

Databases can be hacked through the flaws in their features. Hackers can break into legitimate credentials and compel the system to run any arbitrary code. Although it sounds complex, the access is actually gained through the basic flaws inherent to the features. The database can be protected from third-party access by security testing. Also, the simpler its functional structure, the greater the chances of ensuring effective protection for each database feature.

Weak and Complex DB Infrastructure

Hackers do not generally take control over the entire database in one go. They opt to play a game in which they identify a particular weakness in the infrastructure and exploit it to their advantage. They launch a string of attacks until they finally reach the backend. Security software cannot fully protect your system from such manipulations. Even if you pay attention to the specific feature flaws, it's important not to leave the overall database infrastructure too complex. When it's complex, there's a chance you will forget or neglect to check for and fix its weaknesses. Thus, it is important that every department maintains the same amount of control and segregates systems to decentralise focus and reduce possible risks.

Limitless Administration Access = Poor Data Protection

A clear division of duties between the administrator and the user ensures access is limited to experienced teams. This way, users who are not involved in the database administration process will face greater difficulties if they attempt to steal data. If

you can limit the number of user accounts, it's even better because hackers will face more problems in gaining control over the database as well. This case can be applied to any type of business, but it usually happens in the financial industry. Thus, it's good not only to care about who has access to the sensitive data but also to perform banking software testing before releasing it.

Test Website Security to Avoid SQL Injections

This is a major roadblock to database protection. Injections attack applications, and database administrators are forced to clean up the mess left by malicious code and variables inserted into strings. Web application security testing and firewall implementation are the best options to protect the web-facing databases. However, this is a big problem for online businesses; it's not one of the major mobile security challenges, which is a great advantage for the owners who only have a mobile version of their application.

Inadequate Key Management

It's good if you encrypt sensitive data, but it's also important that you pay attention to who exactly has access to the keys. Since the keys are often stored on someone's hard drive, they are obviously an easy target for anyone who wants to steal them. If you leave such important software security tools unguarded, your system becomes vulnerable to attack.

Irregularities in Databases

It is inconsistencies that lead to vulnerabilities. Test website security and ensure data protection regularly. If any discrepancies are found, they must be fixed as soon as possible. Your developers should be aware of any threat that might affect the database. Though this is not an easy task, proper tracking can keep the information secret.

Despite being aware of the need for security testing, numerous businesses still fail to implement it. Fatal mistakes usually occur during development, but also during app integration or while patching and updating the database. Cybercriminals take advantage of these failures to make a profit, and as a result, your business is at risk of being hacked.